
```
function [x, k, w_opt] = metoda_sor(A, b, w, tol, max_iter)
```

```
n = length(b);  
x = zeros(n, 1);  
x_nou = zeros(n, 1);
```

```
D = diag(diag(A));  
L = tril(A, -1) * -1; % partea strict inferioara  
U = triu(A, 1) * -1;  % partea strict superioara
```

```
% calculul lui w optim  
T_jacobi = D \ (L + U);  
rho = max(abs(eig(T_jacobi)));
```

```
w_opt = 2 / (1 + sqrt(1 - rho^2));
```

```
if isempty(w)  
    w = w_opt;  
end
```

```
M = (1/w)*D - L;  
N = ((1-w)/w)*D + U;
```

```
for k = 1:max_iter
```

```
    x_nou = M \ (N * x + b);
```

```
    if norm(x_nou - x, inf) < tol  
        x = x_nou;  
        return;  
    end
```

```
    x = x_nou;
```

```
end
```

```
warning('metoda SOR nu a convers');
```

```
end
```

```
Not enough input arguments.
```

```
Error in metoda_sor (line 4)
```

```
    n = length(b);  
        ^
```

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